

A Comprehensive Parasitic-Aware Post-Layout Design and Verification Methodology for a 6.5-GHz Ultra-Low-Power Ring Oscillator in 65-nm CMOS

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Abstract—This paper reports the physical implementation and parasitic-aware post-layout characterization of a low-power three-stage CMOS ring oscillator implemented in a 65-nm Low-Power (LP) CMOS process. Operating from a 1.0-V supply, the oscillator is physically implemented using custom layout techniques with a compact active area of $73.96 \mu\text{m}^2$. Following successful design rule checking (DRC), Layout Versus Schematic (LVS) verification, and detailed standard parasitic format (DSPF) extraction, post-layout Periodic Steady State (PSS) and periodic noise (pnoise) analyses are performed to characterize the oscillator under realistic conditions. The extracted design achieves a stable oscillation frequency of 6.50 GHz while consuming only $286.3 \mu\text{W}$ from a 1.0-V supply. The simulated single-sideband phase noise is -77.6 dBc/Hz at a 1-MHz offset, corresponding to a figure of merit (FoM) of -159.3 dBc/Hz . The comparison between schematic and post-layout simulations highlights the significant influence of distributed interconnect parasitics on oscillator performance, emphasizing the importance of parasitic-aware physical verification in nanoscale CMOS circuits. The results provide a concrete reference point for the parasitic budget that must be anticipated when designing low-power multi-gigahertz oscillators in nanoscale CMOS.

Keywords—ring oscillator, CMOS, post-layout verification, parasitic extraction, DSPF, phase noise, low-power design

I. INTRODUCTION

Oscillators are essential building blocks in modern integrated circuits and are widely used in phase-locked loops (PLLs), clock-and-data recovery (CDR) circuits, and other timing systems [1]. Among various oscillator topologies, ring oscillators are attractive because they occupy a small silicon area, do not require inductors, and can be readily implemented

in standard CMOS processes [2], [3]. In particular, the three-stage ring oscillator offers a simple structure and naturally provides multiple phase outputs, making it suitable for many high-speed applications [4], [5].

As CMOS technology scales into the nanometer regime, however, maintaining the accuracy of circuit performance prediction becomes increasingly challenging. During the early design stage, circuit performance is commonly evaluated using schematic-level simulations. While this approach is useful for initial design exploration, it does not account for parasitic elements introduced during physical implementation. At multi-gigahertz frequencies, interconnect resistances and capacitances in the back-end-of-line (BEOL) layers can no longer be neglected. These parasitic components increase signal propagation delay and may cause the actual oscillation frequency to deviate from the intended design value [6].

For this reason, post-layout verification plays an important role in high-frequency circuit design. Although many studies have reported the performance of CMOS ring oscillators, most discussions focus on transistor-level simulations, while the influence of layout parasitics is often treated only briefly. As technology nodes continue to shrink, understanding the difference between schematic and post-layout performance becomes increasingly important for obtaining reliable circuit behavior.

In this work, a low-power three-stage ring oscillator is implemented in a 65-nm Low-Power (LP) CMOS process and evaluated using a complete post-layout design flow. After layout generation, the circuit is verified through design rule checking

(DRC) and layout-versus-schematic (LVS) procedures, followed by RC parasitic extraction in Detailed Standard Parasitic Format (DSPF). The extracted netlist is then analyzed using periodic steady-state (PSS) and periodic noise (pnoise) simulations to characterize the oscillator under realistic operating conditions.

The main contributions of this paper are summarized as follows:

- A node-resolved quantification of BEOL parasitic loading in a 65-nm multi-gigahertz ring oscillator, in which per-net extracted capacitances are reported alongside the resulting 41.7% frequency reduction and 71.2% stage-delay increase.
- A comparison between schematic-level and extracted-netlist simulations to evaluate the effect of BEOL interconnect parasitics on oscillation frequency and propagation delay.
- Post-layout characterization of a compact 6.5-GHz ring oscillator with a power consumption of 286.3 μ W, including phase-noise and figure-of-merit analysis.

II. RING OSCILLATOR OPERATION AND DESIGN METHODOLOGY

A. Topology Selection

A ring oscillator was selected for this study. Compared with LC oscillators, which offer superior phase noise but require a large die area and an on-chip inductor, ring oscillators occupy considerably less area, maintain a relatively constant voltage swing, operate at low supply voltages, and can be realized in any standard CMOS process without additional passive components [2], [3]. These properties suit the 73.96 μm^2 core budget and the 1.0-V supply targeted here.

Beyond area and power, the ring oscillator has a further property that makes it well suited to the purpose of this work: its oscillation frequency is directly and explicitly governed by the propagation delay of its stages, as developed in the following subsection. Any change in node capacitance therefore appears directly as a measurable frequency shift. This makes the ring oscillator an effective vehicle for evaluating switching speed and the influence of design and layout parameters on propagation delay, which is the central concern of this paper.

The three-stage configuration was adopted because it is the minimum odd stage count that satisfies the Barkhausen criterion for a single-ended inverter ring, and has been reported successfully in prior work [5]. A smaller stage count also minimizes the total switched capacitance around the loop, which is consistent with the low-power target.

B. Oscillation Principle and Startup Condition

The proposed oscillator consists of three cascaded CMOS inverter stages connected in a closed loop. Each stage is formed by a PMOS–NMOS pair, and the output of the last stage is fed back to the input of the first stage. Because the loop contains an odd number of inversions, the circuit is able to generate a self-sustained periodic signal.

To maintain oscillation, the loop must satisfy the Barkhausen criterion at the oscillation frequency. This requires the total

phase shift around the loop to reach 360° and the loop gain to be equal to or greater than unity [2]. In a three-stage ring oscillator, the inverter chain contributes an overall phase inversion of 180° , while the frequency-dependent phase shift provided by the stages contributes the remaining 180° required for sustained oscillation.

Assuming that all stages are identical, the oscillation frequency can be related to the dominant pole frequency ω_p of each stage by $\omega_0 = \omega_p \tan(\pi/N)$, which for $N = 3$ gives $\omega_0 = \sqrt{3} \omega_p$. Although this relationship is useful analytically, the oscillation frequency is more conveniently expressed in terms of the propagation delay of each stage. For an N -stage ring oscillator, the oscillation frequency can be approximated by

$$f_{\text{osc}} = \frac{1}{2N\tau_p} \quad (1)$$

where τ_p denotes the average propagation delay of one stage. Since the propagation delay is mainly determined by the effective resistance and load capacitance of the inverter, it can be approximated by

$$\tau_p \approx 0.69 R_{\text{eq}} C_{\text{tot}} \quad (2)$$

where R_{eq} is the equivalent switching resistance and C_{tot} is the total capacitance seen at the output node. Substituting (2) into (1) gives

$$f_{\text{osc}} \approx \frac{1}{1.38 N R_{\text{eq}} C_{\text{tot}}} \quad (3)$$

In practical implementations, the total capacitance consists of both intrinsic load capacitance and parasitic capacitance introduced during physical layout. Therefore,

$$C_{\text{tot}} = C_L + C_{\text{par}} \quad (4)$$

where C_L represents the intrinsic load capacitance and C_{par} denotes the parasitic capacitance associated with interconnect routing and device layout. Consequently, the oscillation frequency can be rewritten as

$$f_{\text{osc}} \approx \frac{1}{1.38 N R_{\text{eq}} (C_L + C_{\text{par}})} \quad (5)$$

Equation (5) shows that additional parasitic capacitance increases the stage delay and reduces the oscillation frequency. This effect becomes increasingly significant at multi-gigahertz frequencies, where interconnect parasitics can no longer be neglected. Therefore, accurate prediction of oscillator performance requires post-layout parasitic extraction and verification, which form the main focus of this work.

C. Short-Channel Propagation Delay Model

Although the startup condition of a ring oscillator can be explained using small-signal analysis, its steady-state oscillation frequency is primarily determined by the propagation delay of the inverter stages. For an N -stage ring oscillator, one complete period corresponds to the time required for a signal transition to propagate twice around the loop. Therefore,

$$T = \frac{1}{f_{\text{osc}}} = 2Nt_d \quad (6)$$

where t_d denotes the average propagation delay of a single stage.

The classical long-channel estimate $t_d \approx C_{\text{node}} V_{\text{DD}} / 2I_{\text{drive}}$ becomes less accurate in 65-nm devices, where velocity saturation and mobility degradation limit the available drain current and the delay is instead set by the saturation current $I_{\text{sat}} = v_{\text{sat}} C_{\text{ox}} W (V_{\text{GS}} - V_{\text{th}} - V_{\text{dsat}})$. In either regime, however, t_d remains proportional to the effective node capacitance, which at the schematic level consists only of intrinsic device capacitance but after physical implementation also includes the interconnect contribution C_{par} of (4). This effect is examined quantitatively through post-layout extraction in the following sections.

D. Pre-Layout Performance Baseline

Fig. 1 shows the schematic of the proposed three-stage CMOS ring oscillator, and Table I summarizes the corresponding device dimensions. To provide a reference for evaluating the impact of layout parasitics, the circuit was first simulated at the schematic level using a supply voltage of $V_{\text{DD}} = 1.0$ V under typical-typical (TT) process conditions at 27°C. All transistors employ a channel length of 60 nm. The NMOS and PMOS devices are sized to $W_n = 2 \mu\text{m}$ and $W_p = 4 \mu\text{m}$, respectively, to provide balanced drive strength.

Under these conditions, the transient simulation produces an oscillation frequency of

$$f_{\text{osc,schematic}} = 11.15 \text{ GHz} \quad (7)$$

which corresponds to a period of

$$T_{\text{schematic}} = \frac{1}{f_{\text{osc,schematic}}} = 89.71 \text{ ps}. \quad (8)$$

Using (6), the average propagation delay of a single inverter stage can be estimated as

$$t_{d,\text{schematic}} = \frac{T_{\text{schematic}}}{2N} = 14.95 \text{ ps} \quad (9)$$

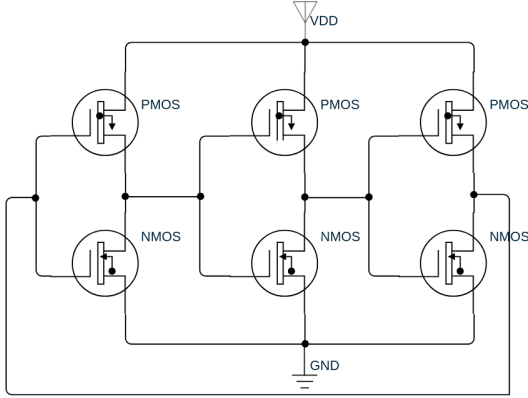


Fig. 1. Schematic of the verified three-stage static CMOS ring oscillator loop.

TABLE I

TRANSISTOR DIMENSIONS OF THE THREE-STAGE RING OSCILLATOR

Device Class	Model	Width (W)	Length (L)
NMOS (3 devices)	nch	$2.0 \mu\text{m}$	60 nm
PMOS (3 devices)	pch	$4.0 \mu\text{m}$	60 nm

for the three-stage configuration ($N = 3$). These results serve as the pre-layout performance baseline and are later used to quantify the influence of interconnect parasitics extracted from the physical layout.

III. PHYSICAL DESIGN AND CUSTOM LAYOUT METHODOLOGY

The physical-design and verification flow followed in this work proceeds from schematic capture to manual layout drawing, DRC, LVS, RC parasitic extraction in DSPF, and finally post-layout PSS and p_{noise} characterization. The layout steps are described in this section; extraction and post-layout simulation are described in Section IV.

A. Device Sizing and Symmetry Considerations

The ring oscillator was implemented in a 65-nm Low-Power (LP) CMOS process. To obtain balanced rise and fall transitions, the PMOS devices were sized twice as wide as the NMOS devices, with $W_p = 4 \mu\text{m}$ and $W_n = 2 \mu\text{m}$. This sizing ratio compensates for the lower hole mobility of PMOS transistors and helps maintain similar pull-up and pull-down drive strengths [7].

All transistors employ the minimum channel length of 60 nm to maximize switching speed and reduce intrinsic delay [2]. The resulting symmetric inverter structure improves transition balance and helps minimize duty-cycle distortion in the oscillation waveform.

B. Floorplanning and Interconnect Routing

A fully custom layout was used to implement the oscillator core, shown in Fig. 2. To simplify routing and maintain symmetry, the three PMOS devices were placed in the upper row within a common n-well, while the corresponding NMOS devices were arranged in the lower row. This arrangement provides a compact structure and reduces unnecessary routing overhead.

The feedback paths between stages were routed using short interconnects primarily in the lower metal layers. Since the oscillator operates in the multi-gigahertz range, minimizing interconnect length is important to reduce parasitic resistance and capacitance [7]. The final layout occupies an area of $8.6 \mu\text{m} \times 8.6 \mu\text{m}$, corresponding to a total core area of $73.96 \mu\text{m}^2$.

C. Matching and Mismatch Considerations

Because the three delay cells nominally set identical stage delays, any systematic difference between them translates directly into duty-cycle distortion and phase error between the outputs. The layout was therefore constrained so that the three stages are drawn as identical instances with identical device orientation and poly pitch, and so that the interstage routing is topologically the same for each cell rather than individually optimized.

The extracted per-net capacitances reported later in Table II indicate that this constraint was met: the two purely internal nodes, *net3* and *net5*, differ by only 0.02 fF out of

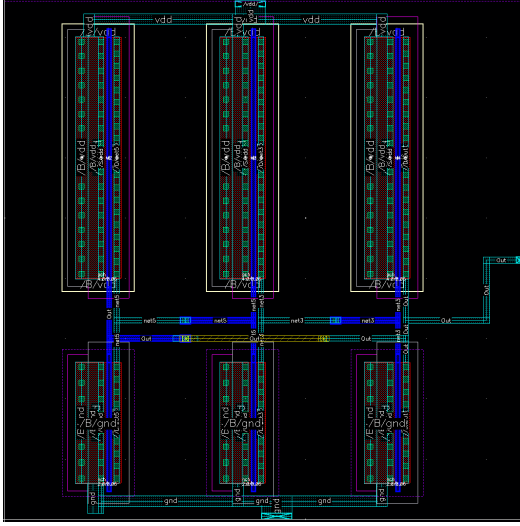


Fig. 2. Manually routed layout core showing row-aligned PMOS and NMOS device matching ($8.6 \mu\text{m} \times 8.6 \mu\text{m}$).

approximately 3.5 fF, i.e. below 1%. The remaining asymmetry is concentrated at the output node, which carries an additional 0.5 fF because it also drives the external observation port. This is a deterministic, layout-induced asymmetry rather than a random-mismatch effect.

Random V_{th} and β mismatch are not captured by the present simulation-only flow. Dummy poly structures at the ends of the transistor rows, continuous n-well and p-substrate guard rings, and matched dummy loading on the internal nodes are identified as the measures required before tape-out, and are noted as future work.

D. Design Rule Checking

After layout generation, Design Rule Checking (DRC) was performed to verify compliance with the design rules of the 65-nm CMOS process. During the initial verification stage, a Metal-1 spacing violation (rule M1.S.1, min $0.09 \mu\text{m}$) was identified and corrected by adjusting the routing geometry. After modification, the layout satisfied all essential DRC requirements. Some remaining messages were associated with density-related rules and chip-level layers (CSR and AP layers), which do not affect the functionality of the oscillator core and are routinely addressed via automated dummy-fill insertion during final mask assembly.

E. Layout-Versus-Schematic Verification

Layout-Versus-Schematic (LVS) verification was carried out to ensure consistency between the physical layout and the original schematic. The comparison includes device types, transistor dimensions, pin assignments, and net connectivity. The extracted layout was found to be consistent with the schematic, indicating that the designed topology was correctly transferred into the physical implementation.

Consequently, the effects observed in the post-layout simulations can be attributed mainly to the parasitic elements introduced by interconnects and device geometries rather than connectivity errors.

IV. PARASITIC EXTRACTION AND POST-LAYOUT SIMULATION METHODOLOGY

A. RC Parasitic Extraction

Following DRC and LVS verification, RC parasitic extraction was performed to generate a Detailed Standard Parasitic Format (DSPF) netlist containing distributed resistance and capacitance associated with interconnects and device geometries [7]. Compared with the schematic-level representation, the extracted netlist provides a more realistic description of the oscillator under practical operating conditions. The resulting per-net capacitances are listed in Table II.

The output node (Out) carries the largest capacitance because of the feedback routing and external observation port, while the two internal nodes are closely matched owing to the symmetrical layout arrangement. These additional components increase the effective node loading and therefore the propagation delay of each stage, so the oscillation frequency obtained from the extracted netlist is lower than that predicted by the ideal schematic model. The influence of these components is analyzed quantitatively in the following section.

B. Post-Layout Simulation Methodology

Post-layout characterization used periodic steady-state (PSS) and periodic noise (pnoise) analyses. Because the extracted DSPF netlist contains many distributed RC elements, the simulation settings were tuned for stable convergence and accurate steady-state solutions. The default `analogLib port` primitive presents a 50Ω source/load resistance. Although this matches standard test instruments, it overloads an unbuffered, low-power sub-micron inverter stage, suppressing the loop gain below the Barkhausen criterion and extinguishing the oscillation. A high-impedance ($1 \text{ M}\Omega$) output port was therefore used to minimize core loading. The PSS analysis ran in oscillator mode with sufficient stabilization time for the amplitude and frequency to settle before the shooting algorithm was applied. For phase noise, `pnoise` was swept over offsets from 1 kHz to 100 MHz using the time-average method with phase-modulation (PM) noise as the output. These settings yielded stable simulations and reliable extraction of oscillation frequency, power consumption, and phase-noise characteristics.

C. Phase Noise and Figure-of-Merit Evaluation

The phase-noise performance of the oscillator was evaluated from the `pnoise` simulations [8], [9]. Single-sideband phase noise was measured at various offset frequencies relative to

TABLE II
EXTRACTED PER-NET PARASITIC CAPACITANCE FROM DSPF

Extracted Net Label	Parasitic Capacitance (fF)
Out	4.07
net3	3.54
net5	3.52
Vdd	3.16
gnd	3.01

the carrier. To facilitate comparison with previously reported oscillators, the figure of merit (FoM) was calculated as [10]

$$\text{FoM} = \mathcal{L}(\Delta f) - 20 \log_{10} \left(\frac{f_0}{\Delta f} \right) + 10 \log_{10} \left(\frac{P}{1 \text{ mW}} \right) \quad (10)$$

where $\mathcal{L}(\Delta f)$ is the single-sideband phase noise at an offset frequency Δf , f_0 denotes the oscillation frequency, and P represents the power consumption.

V. RESULTS AND DISCUSSION

A. Post-Layout Transient Characteristics

The post-layout transient response of the RC-extracted oscillator was characterized under a 1.0-V supply. Fig. 3 shows stable periodic oscillations with nearly rail-to-rail voltage swings ranging from approximately -0.05 V to 1.04 V.

From the steady-state waveform, the oscillation period was measured as

$$T_{\text{post-layout}} = 153.84 \text{ ps} \quad (11)$$

which corresponds to an oscillation frequency of

$$f_{\text{osc,post-layout}} = \frac{1}{T_{\text{post-layout}}} = 6.50 \text{ GHz}. \quad (12)$$

The average supply current obtained from the transient simulation was $286.3 \mu\text{A}$. Consequently, the power consumption of the oscillator is

$$P_{\text{dc}} = V_{\text{DD}} I_{\text{avg}} = 1.0 \text{ V} \times 286.3 \mu\text{A} = 286.3 \mu\text{W}. \quad (13)$$

Using (6), the average propagation delay of each inverter stage is estimated to be

$$t_{\text{d,post-layout}} = \frac{T_{\text{post-layout}}}{2N} = 25.6 \text{ ps}. \quad (14)$$

B. Effect of Layout Parasitics

Table III compares the pre-layout and post-layout performance. The oscillation frequency decreases from 11.15 GHz to 6.50 GHz after parasitic extraction, corresponding to a frequency reduction of approximately 41.7% . In addition, the average stage delay increases from 14.95 ps to 25.6 ps, a 71.2% increase. Because the transistor dimensions (W/L) and

TABLE III
PRE-LAYOUT VERSUS POST-LAYOUT PERFORMANCE

Parameter	Pre-Layout	Post-Layout	Change
Oscillation frequency	11.15 GHz	6.50 GHz	-41.7%
Loop period	89.71 ps	153.84 ps	$+71.5\%$
Stage delay t_d	14.95 ps	25.6 ps	$+71.2\%$

bias conditions remained identical across both setups, this degradation is driven entirely by the distributed RC parasitics embedded in the DSPF netlist.

C. Phase Noise and Figure-of-Merit Performance

Fig. 4 shows the simulated phase-noise spectrum of the oscillator. The curve exhibits a consistent $1/f^2$ slope across the offset spectrum, confirming that white thermal-noise contributions dominate the high-frequency phase-error profile [8], [9].

At an offset frequency of 1 MHz from the 6.50 GHz carrier, the single-sideband phase noise is

$$\mathcal{L}(1 \text{ MHz}) = -77.6 \text{ dBc/Hz}. \quad (15)$$

At an offset frequency of 100 MHz, the phase noise decreases to -124.3 dBc/Hz. Substituting the extracted post-layout metrics ($\mathcal{L}(1 \text{ MHz}) = -77.6$ dBc/Hz, $f_0 = 6.50$ GHz, and $P = 286.3 \mu\text{W}$) into (10) yields

$$\begin{aligned} \text{FoM} &= -77.6 - 20 \log_{10} \left(\frac{6.5 \times 10^9}{1.0 \times 10^6} \right) + 10 \log_{10}(0.2863) \\ &= -77.6 - 76.26 - 5.43 = -159.3 \text{ dBc/Hz}. \end{aligned} \quad (16)$$

The relatively low power consumption of $286.3 \mu\text{W}$ contributes to the obtained FoM value. Although the phase noise is typical of a conventional single-ended ring oscillator, the results demonstrate that the proposed design achieves a reasonable trade-off among oscillation frequency, power consumption, and spectral purity.

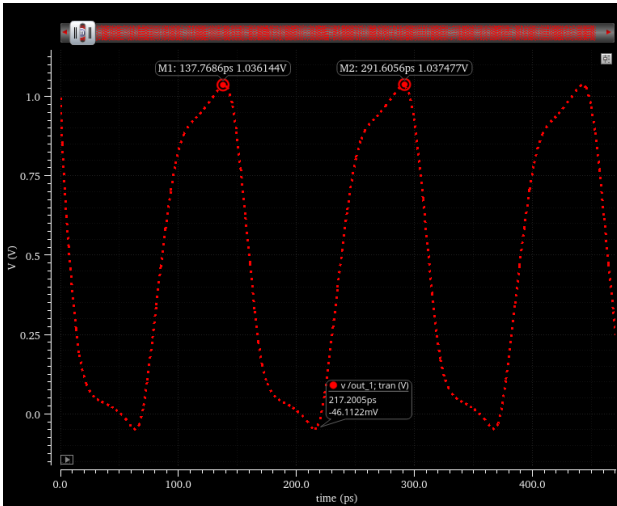


Fig. 3. Post-layout (RC-extracted) transient output waveform showing stable rail-to-rail oscillation at 6.50 GHz with a period of 153.84 ps.

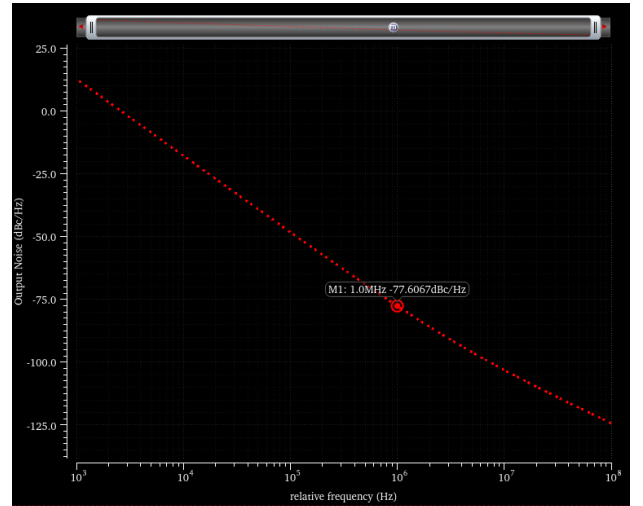


Fig. 4. Post-layout (RC-extracted) single-sideband phase noise versus relative offset frequency, showing -77.6 dBc/Hz at a 1 MHz offset.

TABLE IV
PERFORMANCE COMPARISON WITH REPORTED CMOS RING OSCILLATORS

Parameter	This Work	[5] (2012)	[11] (2017)	[12] (2023)
Technology	65 nm LP	0.18 μm	90 nm	90 nm
Topology	3-st. RO	3-st. CCO	SC-VCO	CS-VCO
Verification	Post-layout	Simulation	Schematic	Sim.
Supply (V)	1.0	1.8	1.8	1.8
Freq. (GHz)	6.50	5.5	4.52–6.02	5.1
Power	286.3 μW	≤ 17.6 mW	0.295 mW	289 μW
Phase noise ^a	−77.6	−104 ^b	N/R	−74
FoM (dBc/Hz)	−159.3	−154.4	−155.5	−152.6
Area (μm^2)	73.96	300 ^c	N/R	N/R

RO: ring oscillator; CCO: current-controlled oscillator; SC-VCO: switched-capacitor ring VCO; CS-VCO: current-starved VCO; Sim.: schematic-level simulation; N/R: not reported. ^aIn dBc/Hz at 1 MHz offset unless noted. ^bReported at 4 MHz offset. ^cCore area 0.0003 mm².

D. Comparison with Previous Works

Table IV compares the proposed oscillator with representative CMOS ring oscillators reported in the literature. As the design is characterized by post-layout simulation, comparison with measured silicon should be treated with caution. Even so, the proposed oscillator attains the best figure of merit (−159.3 dBc/Hz), surpassing the current-controlled RO in [5] (−154.4 dBc/Hz), the switched-capacitor VCO in [11] (−155.5 dBc/Hz), and the current-starved VCO in [12] (−152.6 dBc/Hz). It also uses the lowest supply (1.0 V), the most compact core area (73.96 μm^2), and sub-milliwatt power (286.3 μW) on par with the lowest-power designs. Other topologies, such as the quadrature design in [13], offer additional multiphase outputs at the cost of more delay stages.

Compared with previous studies, the main contribution of this work lies not in introducing a new oscillator topology, nor in the extraction procedure itself, which is a standard step in any modern design flow. It lies in reporting the magnitude of the correction that this flow produces at a specific and fully specified operating point. Had only the schematic-level result been reported, as is common practice, this design would have appeared to operate 4.65 GHz above what the extracted netlist supports. Table III makes that correction explicit, and Table II attributes it to individual nets rather than to a single lumped factor.

VI. CONCLUSION

This paper presented the physical implementation and parasitic-aware post-layout characterization of a low-power three-stage ring oscillator in a 65-nm Low-Power CMOS process. The flow followed here covers custom layout generation, physical verification, RC parasitic extraction, and post-layout characterization using periodic steady-state (PSS) and periodic noise (pnoise) analyses. Each of these steps is well established; what is reported is the quantified effect they reveal.

The extracted oscillator achieved a stable operating frequency of 6.50 GHz while consuming only 286.3 μW from a 1.0-V supply. The phase-noise performance was characterized,

resulting in a single-sideband phase noise of −77.6 dBc/Hz at a 1-MHz offset and a corresponding figure of merit of −159.3 dBc/Hz, occupying a compact core area of 73.96 μm^2 . The comparison between schematic-level and post-layout simulations demonstrates the need to incorporate interconnect parasitics into the design flow. Overall, the reported figures provide a concrete reference point for the parasitic budget that must be anticipated when designing low-power, multi-gigahertz oscillator blocks in advanced CMOS technologies. Against its own schematic baseline, the design loses 41.7% of its oscillation frequency and gains 71.2% in stage delay, driven by 3.5–4.1 fF of routing capacitance per internal node. Future work will focus on experimental silicon validation, a systematic layout optimization performed in two stages (device-core extraction compared against the schematic, followed by interconnect analysis), the addition of explicit frequency-tuning capability, and the application of low-noise delay-cell techniques to further improve phase-noise performance.

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